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Evidence of Inverse Mass Segregation in Dynamically Unevolved Young Open Clusters: A Three-dimensional Analysis Based on Gaia DR3

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Abstract

The primordial spatial distribution of massive stars in star clusters records early fragmentation and assembly processes within molecular clouds. We investigate mass segregation in 871 dynamically unevolved open clusters ($\tau = \mathrm{Age}/t_{\mathrm{rh}} < 1$, where t_{rh} is the half-mass relaxation time) using Gaia DR3 astrometry. To mitigate biases from 2D projection effects, we perform a three-dimensional (3D) reconstruction using Bayesian distance inference. Mass segregation is quantified through the minimum spanning tree ratio, Λ_{MSR}. Our 3D analysis reveals widespread inverse mass segregation, with a median $\Lambda_{\mathrm{MSR}} = 0.72$ for the $N = 5$ most massive stars. Approximately 56% of clusters satisfy the criterion $\Lambda_{\mathrm{MSR}} + \sigma < 1$, where σ is the standard deviation of the random Minimum Spanning Tree length distribution, rising to ∼72% for $N = 20$. In contrast, 2D projected analyses yield a median $\Lambda_{\mathrm{MSR}} = 0.86$ and a detection rate of only ∼10%. The persistence of inverse configurations across the $\tau < 1$ regime suggests that dispersed massive-star distributions may be primordial rather than produced by two-body relaxation. A probabilistic posterior sampling analysis of clusters within 600 pc ($N = 119$) confirms a significant positive correlation between τ and Λ_{MSR} (Spearman $r = 0.252$, $p = 0.006$), providing independent validation. Our results suggest that formation models predicting primordial central concentration may need revision, and highlight the necessity of 3D analysis in cluster dynamics.

Unified Astronomy Thesaurus concepts: Open star clusters (1160); Stellar dynamics (1596); Star counts (1568); Star formation (1569); Gaia (2360)

Materials only available in the online version of record: machine-readable table

1. Introduction

Mass segregation—the preferential concentration of massive stars toward the center of a stellar cluster—is a fundamental observable in stellar dynamics (L. Spitzer 1987). In older, dynamically evolved systems, this spatial configuration is well understood as a natural consequence of two-body relaxation, driving the system toward energy equipartition (S. F. Portegies Zwart et al. 2010). However, the primordial spatial distribution of stars in young, dynamically unevolved clusters remains an open question that continues to be actively investigated (C. J. Lada & E. A. Lada 2003; R. J. Parker & J. E. Dale 2013; S. Dib et al. 2018; V. Coenda et al. 2025).

Theoretical models offer conflicting predictions regarding the initial arrangement of massive stars. Models such as competitive accretion (I. A. Bonnell et al. 1998) and monolithic collapse predict that the most massive stars should preferentially form at the center of the deep gravitational potential well, where gas flows are highly concentrated. Conversely, numerical simulations of hierarchical star formation suggest that stars inherit the substructured, fractal nature of their parental turbulent gas clouds (R. J. Parker et al. 2011; R. J. Parker & J. E. Dale 2013). In these scenarios, clusters can form with massive stars widely distributed across different subclumps, leading to a dispersed or "inverse" mass segregation signature at birth.

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Distinguishing between these theoretical pathways requires large, homogeneous observational samples. The Gaia mission (Gaia Collaboration et al. 2023) has revolutionized this field, enabling the discovery and characterization of thousands of Galactic star clusters (T. Cantat-Gaudin et al. 2018, 2020). Recent multiscale statistical analyses of large Gaia DR3 cluster samples have ruled out the prevalence of primordial mass segregation (S. Zhang 2026). However, the majority of broad statistical studies still rely on two-dimensional (2D) projected positions on the sky. Projection effects can blend centrally located massive stars with foreground or background cluster members, artificially inflating the perceived central concentration and potentially obscuring true inverse segregation signals.

In this paper, we extend the investigation of mass segregation into three-dimensional (3D) space. By focusing exclusively on dynamically unevolved systems ($\tau < 1$) and utilizing Bayesian distance inference to reconstruct 3D cluster geometries, we quantify the true prevalence of inverse mass segregation and systematically evaluate the geometric dilution caused by 2D projection.

2. Data and Sample Selection

2.1. Initial Cluster Catalog

Our analysis is based on the comprehensive open cluster catalog compiled by E. L. Hunt & S. Reffert (2024), which provides high-fidelity membership lists, mean astrometric parameters, and age estimates derived from Gaia DR3. To support the reliability of the structural analysis, we apply a

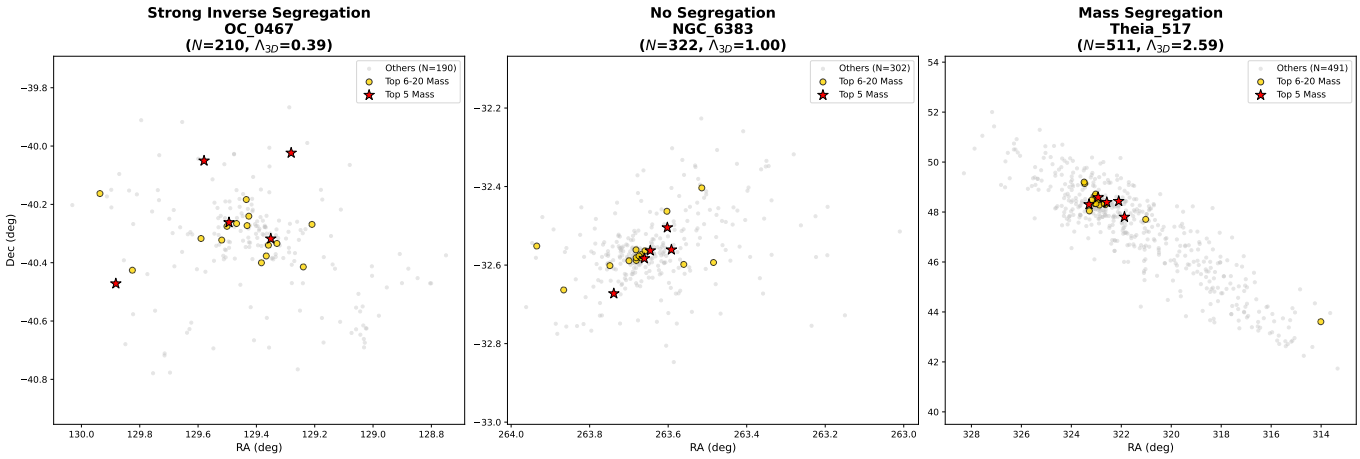


Figure 1. Representative spatial distributions of three open clusters exhibiting distinct mass segregation signatures. Red stars indicate the $N = 5$ most massive stars, gold circles represent the next 15 most massive stars (Top 6–20), and light-gray dots represent all other cluster members. (Left) OC_0467 ($\Lambda_{3D} = 0.39$) shows strong inverse mass segregation, where the most massive stars are widely scattered. (Center) NGC_6383 ($\Lambda_{3D} = 1.00$) shows no significant segregation. (Right) Theia_517 ($\Lambda_{3D} = 2.59$) displays classical, concentrated mass segregation. These examples highlight the diverse initial spatial configurations present in dynamically young clusters.

quality filter, retaining only clusters with at least 50 highly probable members.

2.2. Dynamical Age Criterion

To isolate the primordial or early-assembly dynamical state, we calculate the half-mass relaxation time, $t_{\rm rh}$, for each cluster using the classical formulation (L. Spitzer 1987):

$$t_{\rm rh} = \frac{0.138\, N^{1/2} r_h^{3/2}}{G^{1/2} \bar{m}^{1/2} \ln(0.4N)}, \quad (1)$$

where N is the total number of members, r_h is the half-mass–radius, and $\bar{m}$ is the mean stellar mass.

We adopt the projected half-number radius $r_{\rm 50pc}$ from E. L. Hunt & S. Reffert (2024) as a proxy for the half-mass–radius r_h in the relaxation-time calculation. This is a widely adopted approximation in Gaia-based studies of young clusters, as it is relatively insensitive to the parallax uncertainties in the sparse outer regions that would otherwise introduce substantial scatter in a 3D half-mass–radius estimate. Using the projected radius systematically underestimates the true 3D r_h (due to projection compression), which in turn underestimates $t_{\rm rh}$ and leads to a systematic overestimate of the dynamical age $\tau = {\rm Age}/t_{\rm rh}$ for all clusters in the sample. However, this offset is nearly uniform across the population and does not affect our primary scientific conclusions. The selection criterion $\tau \leqslant 1$ remains conservative, as it effectively excludes clusters that may have begun relaxing even under the underestimated $t_{\rm rh}$. More importantly, the observed inverse mass segregation signals are spatial configurations measured in 3D-reconstructed positions, independent of the absolute value of τ. The persistence of these signals throughout the $\tau < 1$ regime (Figure 5) demonstrates that they are not driven by partial relaxation, regardless of the precise absolute scaling of τ.

We define the dynamical age as $\tau = {\rm Age}/t_{\rm rh}$. By restricting our sample to clusters with $\tau \leqslant 1$, we aim to preferentially select systems in which classical two-body relaxation is unlikely to have substantially altered the primordial spatial configuration. We acknowledge that photometric age determinations for young clusters carry nonnegligible uncertainties, as reflected by differences between age catalogs (L. Cavallo et al. 2024); we note that E. L. Hunt et al. (2025) adopts ages from L. Cavallo et al. (2024) rather than providing an independent determination. The $\tau \leqslant 1$ criterion therefore cannot guarantee a fully unevolved state, but is designed to minimize contamination by dynamically processed systems. This criterion yields a final sample of 871 dynamically unevolved clusters.

To visually illustrate the structural diversity within our unevolved sample, Figure 1 presents the spatial distributions of three representative clusters with varying degrees of mass segregation. While some clusters (e.g., Theia_517) exhibit classical central concentration of their most massive stars, others (e.g., OC_0467) show a strikingly dispersed or "inverse" configuration where massive stars occupy the cluster outskirts. This morphological variety underscores the structural diversity present across the population.

3. Methodology

3.1. Three-dimensional Reconstruction

Uncertainties in parallax measurements (ϖ) can introduce artificial elongation along the line of sight, often referred to as the "Finger-of-God" effect. To mitigate this bias and reconstruct the 3D positions of individual cluster members, we employ a Bayesian distance inference scheme (following guidelines by X. Luri et al. 2018; C. A. L. Bailer-Jones et al. 2021) following the methodology validated in S. Zhang (2026).

The inference is structured as follows. For each star i, the *likelihood* of the observed parallax given the true parallax is modeled as a Gaussian:

$$p(\varpi_{\rm ind}|\varpi_{\rm true}) = \mathcal{N}(\varpi_{\rm true},\ \sigma^2_{\rm ind}), \quad (2)$$

where $\sigma_{\rm ind}$ is the formal Gaia DR3 parallax uncertainty after applying the zero-point correction of L. Lindegren et al. (2021). We adopt a Gaussian cluster-level prior,

$$p(\varpi_{\rm true}) = \mathcal{N}(\varpi_{\rm clus},\ \sigma^2_{\rm clus}), \quad (3)$$

where $\varpi_{\rm clus}$ and $\sigma_{\rm clus}$ are the inverse-variance-weighted mean parallax and its uncertainty derived from all high-probability

members. This prior encodes the physical constraint that cluster members share a common distance to first order. Note that, unlike the Galaxy-wide three-dimensional prior of C. A. L. Bailer-Jones et al. (2021), our cluster-level prior exploits the prior knowledge of cluster membership and is therefore better suited to resolving the internal structure of individual clusters. Because both the likelihood and the prior are Gaussian, the *posterior* is analytically tractable, with mean given by Equation 4 below and posterior variance $\sigma^2_{\rm corr} = (\sigma^{-2}_{\rm ind} + \sigma^{-2}_{\rm clus})^{-1}$.

The corrected parallax is derived based on the astrometric performance of Gaia DR3 (L. Lindegren et al. 2021). Specifically, for each star, a corrected parallax ($\varpi_{\rm corr}$) is derived by combining its individual measurement ($\varpi_{\rm ind}$) with the cluster's collective mean parallax ($\varpi_{\rm clus}$), weighted by their respective uncertainties ($\sigma_{\rm ind}$ and $\sigma_{\rm clus}$):

$$\varpi_{\rm corr} = \frac{\varpi_{\rm ind}\sigma^2_{\rm clus} + \varpi_{\rm clus}\sigma^2_{\rm ind}}{\sigma^2_{\rm clus} + \sigma^2_{\rm ind}}. \quad (4)$$

Uncertainty propagation. The distance is estimated as $d = 1000/\varpi_{\rm corr}$ (pc), with propagated uncertainty $\sigma_d = \sigma_{\rm corr}\, d^2/1000$.

This formulation provides a balanced, mass-symmetric treatment. For massive stars, which are typically brighter and possess smaller $\sigma_{\rm ind}$, the posterior $\varpi_{\rm corr}$ is strongly anchored to the observed $\varpi_{\rm ind}$, faithfully preserving their intrinsic spatial topology. For fainter, lower-mass stars with larger $\sigma_{\rm ind}$, the cluster prior ($\varpi_{\rm clus}$) dominates, effectively suppressing the artificial scattering along the line of sight that would otherwise inflate the apparent cluster volume.

The inferred distance $d = 1000/\varpi_{\rm corr}$ and the sky coordinates (α, δ) are then transformed into a local Cartesian coordinate system (X, Y, Z) centered on the cluster's barycenter. The Z-axis is aligned with the line of sight from the Sun, while the X and Y axes lie in the tangent plane.

As an independent validation of the point-estimate results, we additionally developed a probabilistic posterior sampling approach. For each cluster, $N_{\rm samp} = 50$ independent realizations of stellar positions are drawn from each star's full posterior distribution $\mathcal{N}(\varpi_{\rm corr},\ \sigma^2_{\rm corr})$. This yields a distribution of $\Lambda_{\rm MSR}$ values, from which we extract the median and the probability of inverse segregation $P(\Lambda_{\rm MSR} < 1)$. The probabilistic analysis is applied to a reliability-validated subsample ($d \leqslant 600$ pc, $N = 119$) where the parallax uncertainty is within twice the cluster physical depth, and its results are compared with the full-sample point-estimate analysis in Section 5.

3.2. Quantifying Mass Segregation

Mass segregation is quantified using the Minimum Spanning Tree (MST) method (R. J. Allison et al. 2009, 2010), which measures the geometric compactness of a stellar subset without relying on a predefined cluster center. For a chosen number of the most massive stars, N, the mass segregation ratio is defined as:

$$\Lambda_{\rm MSR} = \frac{\langle L_{\rm random}\rangle}{L_{\rm massive}}, \quad (5)$$

where $L_{\rm massive}$ is the MST length connecting the N massive stars, and $\langle L_{\rm random}\rangle$ is the average MST length of N randomly selected stars, calculated over 200 independent realizations. This number was chosen so that the standard deviation σ of the random MST lengths converges to better than 1% precision (verified by increasing to 2000 realizations in test runs).

A ratio $\Lambda_{\rm MSR} > 1$ indicates classical mass segregation (massive stars are more centrally concentrated), while $\Lambda_{\rm MSR} < 1$ indicates inverse mass segregation (massive stars are more spatially dispersed). To rigorously identify clusters with significant inverse configurations, we adopt the significance threshold:

$$\Lambda_{\rm MSR} + \sigma < 1, \quad (6)$$

where σ is the standard deviation of the random MST lengths ($L_{\rm random}$).

4. Results

4.1. Statistical Distribution of Segregation Signals

The dynamical age distribution of our 871 clusters spans $0.006 < \tau < 0.99$ with a median value of $\tau = 0.43$, confirming their unevolved state. Figure 2 presents the distribution of $\Lambda_{\rm MSR}$ for the $N = 5$ most massive stars. The 3D Bayesian reconstruction reveals a strong tendency toward inverse mass segregation, with a median value of $\Lambda_{\rm MSR} = 0.72$. Applying our significance criterion ($\Lambda_{\rm MSR} + \sigma < 1$), approximately 56% of the clusters exhibit statistically significant dispersed massive-star configurations.

4.2. Dependence on the Massive-star Subsample (N)

To verify that this result is not driven by a few isolated runaway stars, we evaluate the segregation ratio across different values of N. As shown in Figure 3, the fraction of clusters exhibiting significant inverse mass segregation increases steadily with N. For the 3D analysis, this fraction rises from 56% at $N = 5$ to ~72% at $N = 20$. This indicates that the dispersed configuration is a structural characteristic of the upper end of the initial mass function.

We primarily report results for $N = 5$ because, in dynamically unevolved clusters ($\tau < 1$) where classical two-body relaxation has not yet had time to act significantly, any primordial mass segregation (if present) should first manifest in the most massive stars. These stars, being the heaviest tail of the initial mass function, have the shortest local relaxation times and are expected to sink toward the center first if segregation is dynamical or primordial-centralized. Focusing on $N = 5$ thus provides the highest sensitivity to potential early segregation signals while remaining applicable for the majority of our clusters (most have only 5–10 stars more massive than ~5–10 $M_\odot$). Larger values of N (10, 15, 20) were also tested and are shown in Figure 3; they confirm the same global trend toward inverse segregation but with slightly lower sensitivity in sparse clusters due to dilution by intermediate-mass stars. This choice aligns with standard practice in the literature for probing the massive end (R. J. Allison et al. 2009, 2010; R. J. Parker et al. 2011).

4.3. The Dilution Effect of 2D Projection

We systematically compare the 3D segregation ratios with those derived solely from 2D projected coordinates (Figure 4). The 2D analysis yields a substantially weaker signal, with a median $\Lambda_{\rm MSR} = 0.86$, and identifies only ~10% of clusters as

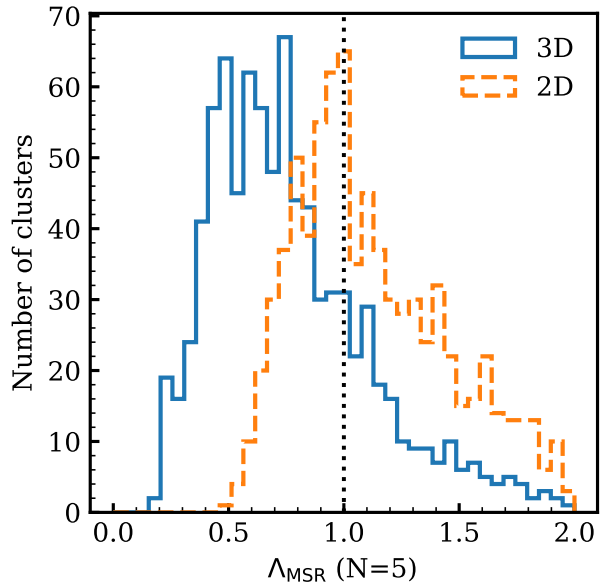


Figure 2. Distribution of the mass-segregation ratio Λ_{MSR} for $N = 5$. The blue histogram shows the results from the 3D Bayesian reconstruction, while the orange histogram represents the 2D projected analysis. The vertical dashed line marks $\Lambda_{MSR} = 1$. The 3D results demonstrate a significant systemic shift toward inverse mass segregation ($\Lambda_{MSR} < 1$).

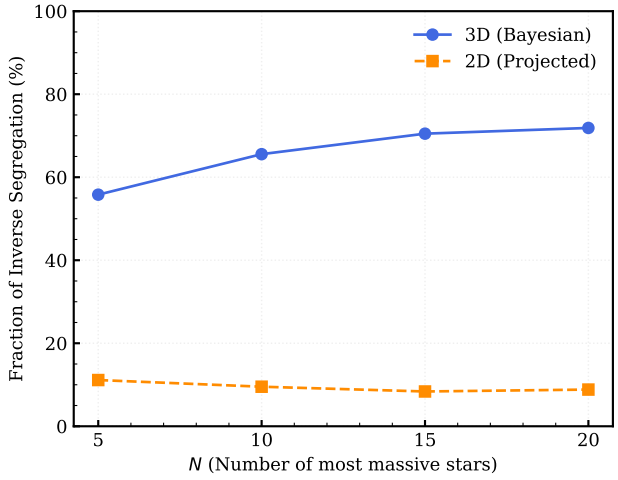


Figure 3. The fraction of clusters satisfying the inverse mass segregation significance criterion ($\Lambda_{MSR} + \sigma < 1$) as a function of the number of massive stars N. The 3D analysis (blue) consistently yields a much higher fraction of inverse segregation across all scales compared to the 2D analysis (orange).

having significant inverse segregation. The majority of data points in Figure 4 fall below the identity line, demonstrating mathematically that geometric projection effects artificially inflate the spatial compactness of massive stars, thereby diluting the true inverse segregation signature.

4.4. Evolution with Dynamical Age

To determine if the inverse segregation is a transient feature rapidly destroyed by early dynamics, we plot Λ_{MSR} against dynamical age τ (Figure 5). The signal persists strongly throughout the full $\tau < 1$ range without a clear evolutionary trend toward $\Lambda_{MSR} = 1$. This stability implies that the dispersed massive-star configuration is a primordial imprint.

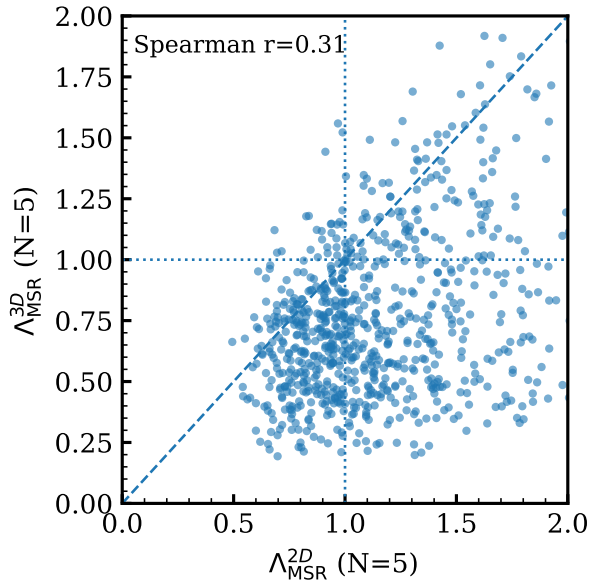


Figure 4. Cluster-by-cluster comparison of Λ_{MSR} ($N = 5$) derived from the 3D reconstruction and the 2D projected analysis. The dashed line indicates equality. The systematic positioning of clusters below the line confirms that projection effects weaken the observed inverse mass segregation.

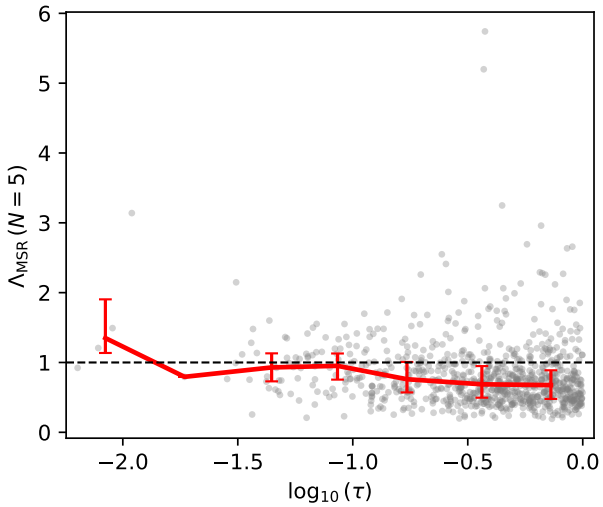


Figure 5. Mass-segregation ratio Λ_{MSR} ($N = 5$) as a function of dynamical age τ = Age/t_{rh}. Gray points represent individual clusters, and the red line connects the median Λ_{MSR} values within seven $\log_{10}(\tau)$ bins. The dashed horizontal line marks $\Lambda_{MSR} = 1$. The inverse configuration is stable across the entire dynamically unevolved regime.

4.5. Representative Cluster Examples

Table 1 lists representative clusters ordered by the mass segregation ratio Λ_5.

5. Discussion and Conclusion

Because our entire sample satisfies $\tau < 1$, classical two-body relaxation cannot account for the observed spatial distributions. The widespread presence of inverse mass segregation in 3D space strongly supports star formation models that predict substantial primordial substructure, where massive stars form in distinct subclumps and remain spatially dispersed during the early-assembly phase (R. J. Parker & J. E. Dale 2013).

Table 1
Representative Clusters Ordered by Λ_5

ID	Name	Λ_5	Λ_{10}	Λ_{15}	Λ_{20}	t_{rh} (Myr)	Age (Myr)	τ
1	OC_0508	0.193	0.308	0.412	0.501	22.68	16.65	0.73
2	HSC_739	0.198	0.393	0.455	0.511	43.96	12.77	0.29
…	…	…	…	…	…	…	…	…
669	Theia_4	0.998	0.730	0.735	0.785	41.24	4.85	0.12
670	FSR_0551	1.001	0.846	0.683	0.688	31.44	6.51	0.21
…	…	…	…	…	…	…	…	…
870	UPK_457	5.199	0.779	0.693	0.702	43.83	16.26	0.37
871	UBC_11	5.741	1.669	1.452	1.334	339.60	127.61	0.38

Note. Clusters are ordered by the mass segregation ratio Λ_5. Only representative systems are shown here. The full catalog of 871 clusters is available in machine-readable format.

(This table is available in its entirety in machine-readable form in the online article.)

Crucially, our Bayesian distance treatment addresses potential concerns regarding over-correction. Because the MST ratio Λ_{MSR} is a relative metric, any symmetric regularization along the line of sight affects both the massive and random subsamples uniformly. In fact, without this constraint, the larger astrometric errors of faint, lower-mass stars would scatter them extensively along the Z-axis, artificially inflating $\langle L_{random} \rangle$ and potentially creating a false-positive inverse segregation signal. By effectively denoising the positions of low-precision members while retaining the high-precision intrinsic spread of massive stars, our method provides a conservative estimate of the true primordial distribution.

To further validate our results, we applied the probabilistic posterior sampling method described in Section 3.1 to clusters within 600 pc ($N = 119$), where the parallax uncertainties are within twice the cluster physical depth. This analysis yields a statistically significant positive correlation between the dynamical age τ and Λ_{MSR} (Spearman $r = 0.252$, $p = 0.006$): the youngest clusters ($\tau \leqslant 0.15$, $N = 43$) exhibit the highest inverse segregation rate of 40%, compared to 22% for older systems ($\tau > 0.15$, $N = 76$; Mann–Whitney $p = 0.015$). This τ-dependence is consistent with two-body relaxation progressively erasing the primordial inverse configuration, and is fully consistent with the full-sample point-estimate analysis, providing independent confirmation of our main conclusion. As a further robustness check, cross-matching with the independent age catalog of L. Cavallo et al. (2024) shows that 74.1% of matched clusters retain $\tau \leqslant 1$, with the median Λ_{MSR} changing by only $\Delta\Lambda = 0.05$, confirming insensitivity to the choice of age catalog.

Our findings differ from some literature reporting positive mass segregation in nearby clusters (S. Dib et al. 2018). We attribute this discrepancy largely to the biases of 2D projected analyses, which mathematically compress spatially distant stars into the same line of sight, masquerading as central concentration. This is consistent with recent large-sample studies that, after rigorous correction for observational biases (including the "Finger-of-God" effect), find no evidence for primordial mass segregation in young open clusters (S. Zhang 2026). In contrast, our focused 3D analysis on the unevolved subsample ($\tau < 1$) reveals a prevalent inverse signature, suggesting that the primordial configuration may be even more dispersed than random in many systems, before dynamical processes begin to act.

Our results align closely with recent N-body simulations (e.g., R. Domínguez et al. 2017) and observations of fractal substructures in young stellar populations (V. Coenda et al. 2025).

In summary, this extensive 3D analysis of Gaia DR3 clusters concludes that:

1. Inverse mass segregation is highly prevalent in dynamically young open clusters, characterizing over half of the solar neighborhood's unevolved systems. An independent probabilistic validation restricted to $d \leqslant 600$ pc ($N = 119$) confirms this finding, yielding 40% significant inverse segregation among the youngest clusters ($\tau \leqslant 0.15$) with a statistically significant τ-dependence ($r = 0.252$, $p = 0.006$).
2. Projection effects severely dilute this signal; thus, 2D analyses systematically underestimate the primordial spatial dispersion of massive stars.
3. Dispersed massive-star distributions are likely established at birth and survive the initial gas expulsion phase.

Future data from Gaia DR4, combined with high-resolution N-body simulations, will provide the kinematic vectors necessary to fully trace how these primordial inverse configurations ultimately evolve into the positively segregated clusters observed at older ages. The improved parallax precision of Gaia DR4 will also substantially expand the volume within which 3D mass segregation analysis is feasible, enabling a definitive test of the τ-dependence identified here.

Acknowledgments

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Software: Astropy (Astropy Collaboration et al. 2022), NumPy (C. R. Harris et al. 2020), SciPy (P. Virtanen et al. 2020), Matplotlib (J. D. Hunter 2007).

Data and Software Availability

The analysis pipeline, processed metrics table, and cluster member data used in this study are permanently archived at Zenodo (v4.0; DOI: 10.5281/zenodo.19177686). The complete Table 1 with all 871 clusters is also provided in CSV format as part of the Zenodo deposit. Users are encouraged to utilize the integrated `Cluster_Analysis_Toolkit.zip` for reproducibility and future analyses. During the preparation of this manuscript, we utilized Python for data processing, Bayesian distance inference, and statistical analysis of the observational data.

AI Disclosure

During the preparation of this work, the authors utilized large language models (LLMs) to assist with English language polishing and proofreading. The authors reviewed, edited, and take full responsibility for the scientific content and integrity of the final manuscript.

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